# Emilie Burnham

PhD Student in Astronomy and Astrophysics | efb5552@psu.edu | GitHub: efburnham | State College, PA

## Research Interests

I am a PhD student in Astronomy and Astrophysics at Penn State University. My research focuses on understanding the star formation histories of galaxies across cosmic time, with a particular emphasis on burstiness. I use a combination of observational data and theoretical modeling to investigate how star formation varies in different populations of galaxies. My work has implications for our understanding of galaxy formation and evolution, as well as the processes that regulate star formation in the universe.

## Education

**PhD, MS** **The Pennsylvania State University**, Astronomy & Astrophysics — State College, PA, US — 2023 – present

- Thesis Advisor: Dr. Joel Leja.
- Expected graduation date: May 2029.

**BSc** **Texas Christian University**, Physics & Astronomy — Fort Worth, TX, US — 2019 – 2022

- Thesis Advisor: Dr. Mia Bovill.
- Graduated summa cum laude.

## Research Experience

**Graduate Research Assistant**, Penn State University — State College, PA, US — 2023 – present

Developed tools (ESPS, PIGLET) to enable fast modeling of galaxy spectra in large surveys. Applied these tools to infer the recent star formation histories of galaxy populations across cosmic time, with a particular emphasis on burstiness. Presented results at conferences and seminars, and published research in peer-reviewed journals.

**NSF DAWN IRES Senior Scholar**, Cosmic Dawn Center (DAWN), Neils Bohr Institute, University of Copenhagen — Copenhagen, Denmark — 2026 – 2026

Mentored undergraduate students participating in the program and led workshops on research skills in galaxy formation and evolution. Continued thesis research while in residence at DAWN.

**NSF REU Research Assistant**, University of Colorado at Boulder — Remote — 2021 – 2021

Conducted research on how well modern radiative transfer codes can recover the absorption features in different spatial features (e.g., plage, faculae) of the solar disk. Presented results at the 2021 Winter American Geophysical Union meeting.

**Undergraduate Research Assistant**, Texas Christian University — Fort Worth, TX, US — 2020 – 2023

Conducted research on modeling the observable affects of a warm dark matter cosmology in the JWST era. Reduced spectra of nearby open clusters to calculate alpha element abundances. Presented results at the 2023 Winter American Astronomical Society meeting.

## Industry Experience

**Clinical Data Analyst**, Front Line Mobile Health, PLLC — Georgetown, TX, US — 2023 – 2024

Conducted research on the health of first responders. Developed a data pipeline to analyze and visualize health data to employers of first responders.

## Teaching Experience

**Undergraduate Research Mentor**, Penn State University — State College, PA — Mar 2026 – present

Students Together for Astronomical Research (STAR) Program. Directly mentored undergraduate student Allyson Garcia on astrophysics research methodologies.

**Guest Lecturer**, Penn State University — State College, PA — Sept 2025 – present

ASTRO 291: Astronomical Methods and the Solar System

**Teaching Assistant**, Texas Christian University — Fort Worth, TX
PHYS 10293: Archeoastronomy — Mar 2023 – present

**Teaching Assistant**, Texas Christian University — Fort Worth, TX
PHYS 20475: Physics I for Majors — Sept 2022 – present

## First-Author Publications

**It's More Complicated Than You Think: A Forward Model to Infer the Recent Star Formation History, Bursty or Not, of Galaxy Populations** — 2026
Burnham, Emilie, Wang, Bingjie, Leja, Joel, Gonzales, Owen, Greene, Jenny E, Iyer, Kartheik G, Mintz, Abby, Setton, David J, Wellons, Sarah, Bezanson, Rachel, others
ui.adsabs.harvard.edu/abs/2026ApJ..1001..205B/abstract (The Astrophysical Journal)

## Co-Author Publications

**Optical Strong Line Ratios Cannot Distinguish between Stellar Populations and Accreting Black Holes at High Ionization Parameters and Low Metallicities** — 2025
Cleri, Nikko J, Olivier, Grace M, Backhaus, Bren E, Leja, Joel, Papovich, Casey, Trump, Jonathan R, Arrabal Haro, Pablo, Buat, Veronique, Burgarella, Denis, Burnham, Emilie, others
ui.adsabs.harvard.edu/abs/2025ApJ...994..146C/abstract (The Astrophysical Journal)

**Population Models for Star Formation Timescales in Early Galaxies: The First Step toward Solving Outshining in Star Formation History Inference** — 2025
Wang, Bingjie, Leja, Joel, Atek, Hakim, Bezanson, Rachel, Burnham, Emilie, Dayal, Pratika, Feldmann, Robert, Greene, Jenny E, Johnson, Benjamin D, Labbe, Ivo, others
ui.adsabs.harvard.edu/abs/2025ApJ...987..184W/abstract (The Astrophysical Journal)

**Hazardous alcohol use and cardiometabolic risk among firefighters** — 2025
Levitt, Danielle E, Wohlgemuth, Kealey J, Burnham, Emilie, Conner, Michael J, Collier, J Jason, Mota, Jacob A
pubmed.ncbi.nlm.nih.gov/39746845 (Alcohol: Clinical and Experimental Research)

**At-risk alcohol use and cardiometabolic risk factors among first responders in the US** — 2024
Levitt, Danielle E, Burnham, Emilie, Conner, Michael J
pmc.ncbi.nlm.nih.gov/articles/PMC11831616 (Drug and Alcohol Dependence)

**Taking a Break at Cosmic Noon: Continuum-selected Low-mass Galaxies Require Long Burst Cycles** — 2026
Mintz, Abby, Setton, David J, Greene, Jenny E, Leja, Joel, Wang, Bingjie, Burnham, Emilie, Suess, Katherine A, Atek, Hakim, Bezanson, Rachel, Brammer, Gabriel, others
ui.adsabs.harvard.edu/abs/2026ApJ...998..257M/abstract (The Astrophysical Journal)

**Occupational Health Disparities: A Profile of Firefighters and Police Officers** — 2026
Wohlgemuth, Kealey J, Conner, Michael J, Burnham, Emilie, Mota, Jacob A
pubmed.ncbi.nlm.nih.gov/41460691 (The Journal of Strength and Conditioning Research)

**UNCOVER/MegaScience Finds Uniform and Highly Bursty Star Formation at 3< z< 9, consistent with the High-Redshift UV Luminosity Function** — 2026
Mitsuhashi, Ikki, Suess, Katherine A, Leja, Joel, Bezanson, Rachel, Greene, Jenny E, Burnham, Emilie, Khullar, Gourav, Mintz, Abby, Nanayakkara, Themiya, Glazebrook, Karl, others
ui.adsabs.harvard.edu/abs/2026arXiv260116284M/abstract (arXiv preprint arXiv:2601.16284)

**RUBIES: The Evolution of the Ionization Parameter from 0< z< 9** — 2026
Cleri, Nikko J, Lewis, Zach J, Leja, Joel, Helton, Jakob M, Burnham, Emilie, Curtis, Olivia, de Graaff, Anna, Hirschmann, Michaela, Katz, Harley, Maseda, Michael V, others
ui.adsabs.harvard.edu/abs/2026arXiv260530410C/abstract (arXiv preprint arXiv:2605.30410)

## Awards

**Harold and Nancy O'Connor Graduate Excellence Fund Scholarship** — 2025

Penn State Eberly College of Science. Eberly Research Showcase. First-place award at a student/postdoc-led research event featuring talks and posters judged by undergraduates, graduate students, postdocs, and faculty on communication effectiveness, with participation from all departments in the Eberly College of Science.

**Frymoyer Award** 2025

Institute of Gravitation and the Cosmos. Awarded to IGC students who have achieved superior academic records or who manifest promise of outstanding academic success. The evaluation of the nominees was based on the nomination letters submitted by faculty members and the packets of materials submitted by the nominees.

**Zaccheus Daniel Foundation for Astronomical Science Scholarship** 2025

Penn State Department of Astronomy and Astrophysics. Result of major contributions and achievements as a graduate student in the Penn State Astronomy & Astrophysics Department. Students were nominated by the faculty, and all nominees were evaluated by a committee based on criteria that included academic merit, research accomplishments, and service to the department.

**University Graduate Fellowship** 2024

Penn State University. Intended to support recruitment of the very highest caliber prospective students to Penn State's doctoral research programs. The prospects are individuals whose goals are to engage in original research and contribute new knowledge to their perspective fields for the ultimate benefit of society.

**Stephen B. Brumbach Distinguished Graduate Fellowship in Science** 2023

Penn State University

**Verne M. Willaman Distinguished Graduate Fellowship in Science** 2023

Penn State University

**Homer F. Braddock Scholarship in Biology, Chemistry, and Physics** 2023

Penn State University

**Dr. C.A. Quarles Physics and Astronomy Scholarship** 2021

Texas Christian University Department of Physics and Astronomy. Awardees are strong candidates for acceptance to graduate school. Received the award twice in separate application cycles. Two-time awardee.

**Joseph Morgan Physics Scholarship** 2021

Texas Christian University Department of Physics and Astronomy. One scholarship per year, this award is based on the recommendation of the faculty for students who have demonstrated the greatest excellence and achievement.

**Bob Bolen Leadership Scholarship** 2021

The Sarah & Ross Perot Jr. Foundation, Texas Christian University. Texas Christian University. Awardees exhibit leadership ability and/or potential and a commitment to community service.

**TCU Scholar** 2021

Texas Christian University. Maintained a 4.0 GPA. Three-time awardee.

**College of Science and Engineering Dean's List** 2019

Texas Christian University College of Science and Engineering. Six-time awardee.. Texas Christian University College of Science and Engineering

## Leadership / Service

**Astronomy on Tap State College Organizer**, Astronomy on Tap State College — State College, PA, 2024 – present

Organized monthly public talks on astronomy topics, coordinated with speakers, and promoted events to engage the local community in astronomy outreach.

**Society of Physics Students (SPS) President and Treasurer**, Texas Christian University Chapter of the Society of Physics Students — Fort Worth, TX, 2021 – 2022

Led the student chapter of the Society of Physics Students, organized events and meetings, managed finances, and coordinated events to grow participation across departmennts.

## Outreach

**Public Lecture on "Why is Astronomy a Big Data Field?"**, Astronomy on Tap State College — State College, PA, 2025

Presented a public lecture on the role of data science in modern astronomy, highlighting how astronomers use large datasets to understand the universe and the importance of data analysis skills in the field.

**Public Lecture on "The Milky Way Through the Rainbow"**, Astro Night at Penn State — State College, PA, 2024

Presented a public lecture on how astronomers use different wavelengths of light to study the Milky Way galaxy, showcasing the diverse phenomena that can be observed across the electromagnetic spectrum.

**Public Lecture on "Suprises at the Edge of the Observable Universe with JWST"**, Central Pennsylvania Observer's Society — State College, PA, 2024

Presented work on the latest discoveries from the JWST, focusing on unexpected findings about galaxies in the early universe and their implications for our understanding of cosmic history.

## Professional Presentations

- **Invited Talks**
  - "It's More Complicated Than You Think: Inferring the Recent Star Formation History, Bursty or Not, in Galaxy Populations"
    - Penn State University Center for Astrostatistics and Astroinformatics Lunch Seminar | State College, PA | 2026
    - University of Copenhagen Cosmic Dawn Center (DAWN) Cake Talk Seminar | Copenhagen, Denmark | 2026
    - University of Pittsburgh Joint Extragalactic Group Meeting | Pittsburgh, PA | 2025
    - Penn State University Institute for Gravitation and the Cosmos Primordial Universe and Gravitation Seminar | State College, PA | 2025
    - Princeton University Galactic/Extragalactic Reading Group | Princeton, NJ | 2025
    - Presented work on inferring and modelingstar formation timescales across galaxy populations.
- **Contributed Talks**
  - "It's More Complicated Than You Think: Inferring the Recent Star Formation History, Bursty or Not, in Galaxy Populations"
    - Eberly Research Showcase | State College, PA | 2025 | First-place prize
    - Penn State University Department of Astronomy and Astrophysics Lunch Talk Seminar | State College, PA | 2025
    - 7th Neighborhood Workshop on Astrophysics and Cosmology | State College, PA | 2025 | First-place prize
    - Presented work on inferring star formation timescales across galaxy populations.
  - "The Need for Speed in Galaxy Evolution Surveys"
    - Penn State Astronomy and Astrophysics Lunch Talk Seminar | State College, PA | 2025
    - Presented work on emulating stellar population synthesis models to enable fast modeling of spectra in large galaxy surveys.
  - "Measuring the Family Histories of Bursty Star Formation within Galaxy Populations"
    - Penn State Astronomy and Astrophysics Lunch Talk Seminar | State College, PA | 2024
    - Presented work on modeling star formation timescales and the resulting observables in bursty galaxy populations.
  - "Forensic Astronomy"
    - Texas Christian University Annual Student Research Symposium | Fort Worth, TX | 2021
    - Evaluated early cluster composition metrics.
- **Contributed Posters**
  - "It's More Complicated Than You Think: Inferring the Recent Star Formation History, Bursty or Not, in Galaxy Populations"
    - Carnegie Mellon University Statistical Methods for the Physical Sciences Research Center Workshop | Pittsburgh, PA | 2025 | simulation-based inference

  - Penn State University Institute for Computational Data Sciences Symposium | State College, PA | 2025 | computational methods
  - Presented work on inferring star formation timescales across galaxy populations.
- "Warm or Cold Dark Matter? A Love-Heat Relationship"
  - Texas Christian University Annual Student Research Symposium | Fort Worth, TX | 2022
  - American Astronomical Society Winter Meeting | Seattle, WA | 2023
  - Presented work on modeling the observable affects of a warm dark matter cosmology in the JWST era.
- "Spectral Runway: An Analysis of Solar Balmer Lines through Observations and Models"
  - American Geophysical Union Winter Meeting | New Orleans, LA | 2021

## Skills

**Computing:** Python, JAX, Machine Learning, Data Analysis, Data Visualization, Software Development, High-Performance Computing, Git, Cloud Computing, GPUs